

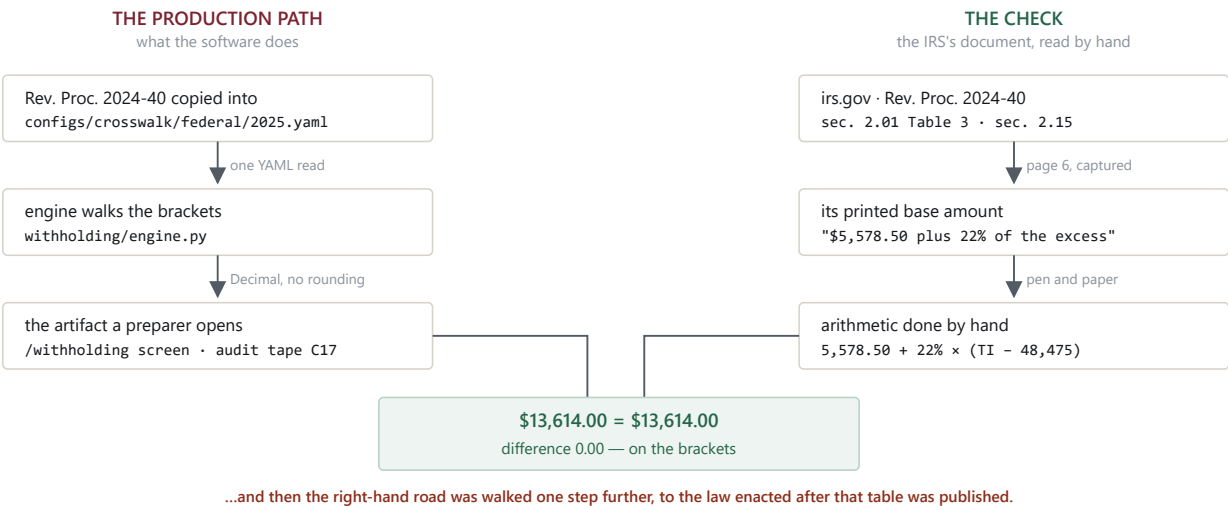
Does the withholding estimator agree with the IRS?

One figure — the federal tax on a single filer with \$100,000 of wages for 2025 — traced from the screen a preparer reads, back to the Internal Revenue Service's own published tables. Nothing here is confirmed by SATC's own software.

6 September 2026 · satc_system @ main 70144b9 · engine src/satc/withholding/engine.py

HOW THE DOTS CONNECT

Two roads carry the same fact. The left is what the software does. The right is the IRS's own document, read by hand. They are supposed to meet.



Only the right-hand road touches a document SATC does not control. That is the road that makes this evidence rather than a second opinion from the same source.

THE ROSTER

Tied out: 19 of 20 checks		
DIFFERS	1	the 2025 standard deduction is superseded by P.L. 119-21 (link 6)
TIED	19	18 printed bracket bases + 1 full computed figure
COULD NOT	0	

The one that differs is below the ones that agree, because the nineteen that agree are not what anybody needs to read.

LINK 1 • THE FIGURE, READ OUT OF THE ARTIFACT

The number being proved is **Total tax liability \$13,614** for the case below. It was read out of **two** artifacts a person actually opens — not from the engine's return value.

ARTIFACT	HOW THE VALUE WAS READ	VALUE
The /withholding screen	Opened in Chrome at <code>127.0.0.1:5095/withholding</code> , form submitted, row Total tax liability in the Result panel	\$13,614
The audit tape workbook	<code>audit-tape-single-100k-2025.xlsx</code> , sheet Withholding Estimate , cell C25 (label in A25)	13614

Standard deduction	\$15,000
Taxable income	\$85,000
Ordinary income tax	\$13,614
Capital-gains tax	\$0
Total tax liability	\$13,614
Marginal / effective rate	22.0% / 13.6%

The Result panel, enlarged. **Ordinary income tax \$13,614** ringed; **Total tax liability \$13,614** beneath it. Read off the screen, not recomputed.

YTD taxable wages
YTD federal tax withheld

Household

Filing status

Single

Tax year

2025

Other income (annual; leave blank if none)

Interest

Qualified dividends

Short-term capital gains

Ordinary dividends

Long-term capital gains

Self-employment net

Deductions, credits & goals

Itemized deductions (blank = standard)

Child tax credit

Estimated tax payments

Prior-year tax (safe harbor)

Extra standard deductions (65+ /blind)

Other nonrefundable credits

Target refund

Prior-year AGI

Run estimate

Result — TY2025

Download audit tape (t

Projection (all jobs)

Projected taxable wages	\$100,000
Adjusted gross income	\$100,000
Standard deduction	\$15,000
Taxable income	\$85,000
Ordinary income tax	\$13,614
Capital-gains tax	\$0
Total tax liability	\$13,614
Marginal / effective rate	22.0% / 13.6%

Recommendation

Projected total payments	
Projected balance	\$13,6
Additional per period (W-4 4c)	\$13,1
On track / under target.	
Tuning Tie-out case; 1 pay periods remaining.	

The same screen in full, so the figure is visible in its context — standard deduction \$15,000, taxable income \$85,000, and the inputs above it.

LINK 2 • THE CALL

Run the app against a scratch store, then post the case. Both blocks are copy-pastable with the real values in them.

```
cd C:/Users/ajish/Documents/Main/Claude/SATC_Prod_Software/SATC/satc_system
SATC_DATA_DIR=/tmp/tieout PYTHONPATH=src .venv/Scripts/python.exe -c "from satc.app.server import create_app;
create_app().run(port=5095)"
```

```
curl -s -X POST http://127.0.0.1:5095/withholding \
-d filing_status=single -d tax_year=2025 \
-d j0_name=Tie-out+case -d j0_pay_frequency=annual \
-d j0_gross_pay_per_period=100000 \
-d j0_federal_tax_withheld_per_period=0 \
-d j0_retirement_pretax_per_period=0 \
-d j0_pay_periods_remaining=1
```

LINK 3 • THE DERIVATION, SO A PERSON CAN REDO IT

Not "then the system computes it". Every step, in order:

STEP	WHAT HAPPENS	RUNNING FIGURE
1	Annual gross wages, one pay period, nothing pre-tax	100,000.00
2	No other income, no above-the-line adjustments → AGI	100,000.00
3	Subtract the standard deduction for a single filer	-15,000.00
4	Taxable income	85,000.00
5	85,000 falls in the band "over \$48,475, not over \$103,350"	—
6	Base for that band	5,578.50
7	$22\% \times (85,000 - 48,475) = 22\% \times 36,525$	8,035.50
8	No capital gains, no SE tax, no NIIT, no credits	0.00
9	Total tax liability	13,614.00

All arithmetic is `Decimal`; the engine never rounds between steps.

The Internal Revenue Service publishes the 2025 rate schedules in **Rev. Proc. 2024-40**. Anyone can fetch it:

```
https://www.irs.gov/pub/irs-drop/rp-24-40.pdf
```

DOCUMENT	SECTION	TABLE	PAGE
Rev. Proc. 2024-40	2.01 Tax Rate Tables	3 – § 1(j)(2)(C) Unmarried Individuals	6
ROW	COLUMN	UNITS	FETCHED
Over \$48,475 but not over \$103,350	The Tax Is	whole dollars, not thousands	6 Sep 2026

If Taxable Income Is:	The Tax Is:
Not over \$11,925	10% of the taxable income
Over \$11,925 but not over \$48,475	\$1,192.50 plus 12% of the excess over \$11,925
Over \$48,475 but not over \$103,350	\$5,578.50 plus 22% of the excess over \$48,475
Over \$103,350 but	\$17,651 plus 24% of

The row this figure depends on, enlarged and ringed: **\$5,578.50 plus 22% of the excess over \$48,475.**

6	
TABLE 3 - Section 1(j)(2)(C) – Unmarried Individuals (other than Surviving Spouses and Heads of Households)	
If Taxable Income Is:	The Tax Is:
Not over \$11,925	10% of the taxable income
Over \$11,925 but not over \$48,475	\$1,192.50 plus 12% of the excess over \$11,925
Over \$48,475 but not over \$103,350	\$5,578.50 plus 22% of the excess over \$48,475
Over \$103,350 but not over \$197,300	\$17,651 plus 24% of the excess over \$103,350
Over \$197,300 but not over \$250,525	\$40,199 plus 32% of the excess over \$197,300
Over \$250,525 but not over \$626,350	\$57,231 plus 35% of the excess over \$250,525
Over \$626,350	\$188,769.75 plus 37% of the excess over \$626,350
TABLE 4 - Section 1(j)(2)(D) – Married Individuals Filing Separate Returns	
If Taxable Income Is:	The Tax Is:
Not over \$11,925	10% of the taxable income
Over \$11,925 but not over \$48,475	\$1,192.50 plus 12% of the excess over \$11,925
Over \$48,475 but not over \$103,350	\$5,578.50 plus 22% of the excess over \$48,475
Over \$103,350 but not over \$197,300	\$17,651 plus 24% of the excess over \$103,350
Over \$197,300 but not over \$250,525	\$40,199 plus 32% of the excess over \$197,300
Over \$250,525 but not over \$375,800	\$57,231 plus 35% of the excess over \$250,525
Over \$375,800	\$101,077.25 plus 37% of the excess over \$375,800
7	
TABLE 5 - Section 1(j)(2)(E) – Estates and Trusts	
If Taxable Income Is:	The Tax Is:
Not over \$3,150	10% of the taxable income
Over \$3,150 but not over \$11,450	\$315 plus 24% of the excess over \$3,150
Over \$11,450 but not over \$15,650	\$2,307 plus 35% of the excess over \$11,450
Over \$15,650	\$3,777 plus 37% of the excess over \$15,650

Page 6 as served by irs.gov, captured in Chrome — the whole of Table 3 (Single) and Table 4 (MFS), so the ringed row can be seen in place. The page number and table headings are in the same shot as the figures.

Read twice. Every figure above was taken from the rendered page *and* from the same PDF's embedded text layer, extracted with `pypdf`. The two readings agree. One apparent disagreement resolved on inspection and is recorded under *what I got wrong*.

LINK 5 • THE COMPARISON

VERDICT • TIED

ours	(/withholding screen, Result → Total tax liability)	13,614.00
ours	(audit tape .xlsx, sheet "Withholding Estimate", C25)	13,614.00
source	(IRS Rev. Proc. 2024-40 §2.01 Table 3, p.6, by hand)	
	$5,578.50 + 0.22 \times (85,000.00 - 48,475.00)$	13,614.00
diff		0.00

The four sameness tests, each checked rather than assumed:

TEST	SAME?	HOW IT WAS CHECKED
Entity	yes	Both sides are a single filer, no dependants — a synthetic case, no client data
Period	yes	Tax year 2025 on both sides; the screen prints "Result — TY2025"
Basis	yes	Federal ordinary income tax before credits and before payroll taxes, on both sides
Units	yes	Whole US dollars on both; the Rev. Proc. tables are not in thousands

THE BRACKET STRUCTURE, ALL OF IT

One figure proves one path through the table. The IRS prints, for every band, the tax accumulated below it — \$5,578.50 , \$17,651 , and so on. Those printed bases are an independent arithmetic statement about the whole schedule, so each one can be checked against what SATC's own bracket edges imply. Eighteen of them, across three filing statuses:

--- single ---	IRS printed	ours implies	diff
11,925	1,192.50	1,192.50	0.00
48,475	5,578.50	5,578.50	0.00
103,350	17,651.00	17,651.00	0.00
197,300	40,199.00	40,199.00	0.00
250,525	57,231.00	57,231.00	0.00
626,350	188,769.75	188,769.75	0.00
--- married filing jointly ---			
23,850	2,385.00	2,385.00	0.00
96,950	11,157.00	11,157.00	0.00
206,700	35,302.00	35,302.00	0.00
394,600	80,398.00	80,398.00	0.00
501,050	114,462.00	114,462.00	0.00
751,600	202,154.50	202,154.50	0.00
--- head of household ---			
17,000	1,700.00	1,700.00	0.00
64,850	7,442.00	7,442.00	0.00
103,350	15,912.00	15,912.00	0.00
197,300	38,460.00	38,460.00	0.00
250,500	55,484.00	55,484.00	0.00
626,350	187,031.50	187,031.50	0.00
VERDICT: TIED - all 18 printed bases reproduced exactly			

This is a check that could have failed. A single mistyped bracket edge shifts every base above it, and the IRS's printed figure would not match.

Rev. Proc. 2024-40 is the authority for the 2025 *inflation adjustments*, published October 2024. It is not the last word on 2025 law. SATC's own crosswalk file says so, in a note nobody had acted on:

```
configs/crosswalk/federal/2025.yaml, line 10

NOTE: several TCJA provisions were later modified by P.L. 119-21
(OBBBA, July 2025); reconcile to enacted law before filing TY2025.
```

Public Law 119-21 — the One Big Beautiful Bill Act, signed 4 July 2025 — raised the standard deduction **for tax year 2025 itself**, not for 2026 onwards. The IRS publishes the enacted figures on a page written specifically for people adjusting their withholding mid-year:

An increase in the standard deduction. The standard deduction has increased to the following amounts:

- \$31,500 (up from \$30,000) for married filing jointly or a qualifying surviving spouse;
- \$23,625 (up from \$22,500) for head of household; and
- \$15,750 (up from \$15,000) for single or married filing separately.

Enlarged. Each line states the enacted figure **and the Rev. Proc. figure it replaced** — and the one it replaced is the one SATC is still using.

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being over 2,000 pounds and that has undergone final assembly in the United States. The deduction is available for both itemizing and non-itemizing taxpayers. The deduction phases out for taxpayers with modified adjusted gross income over \$100,000 (\$200,000 for joint filers). The taxpayer must include the vehicle identification number (VIN) of the applicable passenger vehicle on the tax return for any year in which the deduction is claimed.

The location of final assembly will be listed on the vehicle information label attached to each vehicle on a dealer's premises. Alternatively, taxpayers may rely on the vehicle's plant of manufacture as reported in the VIN to determine whether a vehicle has undergone final assembly in the United States. The [VIN Decoder](#) website for the National Highway Traffic Safety Administration (NHTSA) provides plant of manufacture information. Taxpayers can follow the instructions on that website to determine if the vehicle's plant of manufacture was located in the United States.

Enhanced deduction for seniors. Individuals who are age 65 and older may claim an additional deduction of \$6,000 (\$12,000 for married couples if both spouses qualify). This new deduction is a personal exemption that is separate from the current additional standard deduction for seniors under existing law. The deduction is available for both itemizing and non-itemizing taxpayers. The deduction phases out for taxpayers with modified adjusted gross income over \$75,000 (\$150,000 for joint filers). Each taxpayer claiming the additional deduction must include his or her social security number valid for employment on the return, and, if the taxpayer is married, the taxpayer and the taxpayer's spouse must file jointly.

An increase in the standard deduction. The standard deduction has increased to the following amounts:

- \$31,500 (up from \$30,000) for married filing jointly or a qualifying surviving spouse;
- \$23,625 (up from \$22,500) for head of household; and
- \$15,750 (up from \$15,000) for single or married filing separately.

An increase to the limitation on the itemized deduction for state and local taxes. Individuals who itemize their deductions can claim up to \$40,000 (\$20,000 if married filing separately) for state and local taxes paid. This limit was increased from \$10,000 (\$5,000 if married filing separately). The maximum deduction is reduced for taxpayers with modified adjusted gross income over \$500,000 (\$250,000 if married filing separately).

Changes to the child tax credit. The maximum child tax credit has increased to \$2,200 (up from \$2,000) per eligible child. The individual claiming the child tax credit must have a social security number valid for employment. If married filing jointly, at least one spouse must have a social security number valid for employment. The qualifying child must also have a social security number valid for employment.

Page Last Reviewed or Updated: 28-Jul-2026

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The three enacted amounts, ringed. The IRS states each as an increase from the Rev. Proc. figure — **\$15,750 (up from \$15,000)** — which is precisely the figure SATC is still using. The page's own "Last Reviewed or Updated" date is in the same shot.

VERDICT • DIFFERS

	ours (crosswalk)	IRS (P.L. 119-21)	diff
single / MFS	15,000.00	15,750.00	-750.00
married filing jointly	30,000.00	31,500.00	-1,500.00
head of household	22,500.00	23,625.00	-1,125.00

What it costs on the figure this exhibit set out to prove. The engine's arithmetic is not at fault — hand it the enacted deduction and it lands on the IRS schedule to the cent:

	deduction	taxable inc	tax
as the software ships today	15,000.00	85,000.00	13,614.00
with the enacted OBBBA figure	15,750.00	84,250.00	13,449.00
		OVERSTATED BY	165.00

hand, from IRS Table 3:	$5,578.50 + 22\% \times (84,250 - 48,475)$	$= 13,449.00$
engine, given the enacted figure		$= 13,449.00$
diff		0.00

So the estimator would tell this taxpayer to withhold **\$165 more** than 2025 law requires. On a household, the joint figure is \$1,500 of deduction — and the error grows with the marginal rate.

Which side is believed: the IRS. A statute enacted in July 2025 governs tax year 2025 over a revenue procedure published in October 2024. This is not a close call, and it is not being plugged or rounded away — the crosswalk needs the enacted figures, and the note in that file needs deleting only once they are in.

WHAT IT FOUND

- **The estimator overstates 2025 federal tax by \$165 on the case above, and by more on a joint return.** It uses the standard deduction as published in October 2024, superseded by law in July 2025. Nothing in the test suite could catch this: the tests hand-compute their expected figures from the same tables the engine reads, so the code and the tests agree with each other and both are wrong.
- **The file knew.** `2025.yaml` carries a written instruction to reconcile to enacted law before filing, and it had not been done. A note in a config that nothing enforces is a note that gets read once.
- **2026 is worse and says so.** `load_tax_tables(2026)` returns 2025's tables with the note "*Federal tax tables for 2026 are not fully published in the crosswalk; using 2025 tables for this estimate.*" Honest, and unusable for a 2026 estimate.
- **The bracket machinery is sound.** Eighteen printed base amounts across three filing statuses reproduce exactly, and the engine's arithmetic given a correct deduction lands on the IRS schedule to the cent. The defect is one constant, not the design.

WHAT I GOT WRONG

- **I nearly reported a discrepancy that was not one.** The text extracted from page 5 showed a top-bracket base of `$187,031.50` where the rendered page 6 showed `$188,769.75`. I read that as the two renderings disagreeing. They were different tables — Heads of Households and Single, both topping out at \$626,350. The wider HOH bands accumulate less tax, so the lower figure is right where it sits. Checked before it was written down, and recorded because a tie-out that hides its own misreads is worth nothing.
- **My first framing of this exercise was too narrow.** I set out to prove the estimator against Rev. Proc. 2024-40 and would have stopped at *TIED, difference 0*. That verdict is true and would have been misleading — the revenue procedure is not the same thing as enacted law, and the interesting question was one step past the source I had chosen.

WHAT THIS DOES NOT PROVE

- **One path through the engine.** Wages, standard deduction, one bracket band. **Not tied to any IRS source:** the capital-gains stacking, the self-employment tax, the Additional Medicare Tax, the Net Investment Income Tax, the safe-harbor calculation, or the per-paycheck W-4 line 4c recommendation. Each has a unit test; each of those tests computes its expected value from the same constants the engine uses.

- **Federal only.** Ohio and every other state withholding is outside the estimator entirely, and most SATC clients file one.
- **The other constants in the crosswalk.** Eighteen bracket bases and three standard deductions were checked against the IRS. The Social Security wage base, the SE tax rates, the NIIT and Additional Medicare thresholds, and the capital-gains breakpoints were **not**.
- **The paystub reader.** This exhibit typed its inputs. Nothing here tests whether a scanned paystub yields the right figures to feed it.
- **The full OBBBA reconciliation.** This found one superseded constant by following one note. P.L. 119-21 also created deductions for tips, overtime, vehicle loan interest and seniors, raised the SALT cap, and changed the child tax credit — **none of which the estimator models.** The IRS's own page says its Tax Withholding Estimator is not updated for most of them either. How much that matters has not been assessed.

HOW TO RUN THIS YOURSELF

Every step, with the real values in it. No editing required.

```
cd C:/Users/ajish/Documents/Main/Claude/SATC_Prod_Software/SATC/satc_system
```

```
PYTHONPATH=src .venv/Scripts/python.exe -c "from decimal import Decimal as D; from satc.withholding.engine
import estimate; from satc.withholding.models import EstimatorInput;
r=estimate(EstimatorInput.from_dict({'filing_status':'single','tax_year':2025,'paystub':
{'pay_frequency':'annual','taxable_wages_per_period':100000,'pay_periods_remaining':1})));
print('deduction',r.breakdown.deduction_used,'| taxable',r.breakdown.taxable_income,'|
tax',r.breakdown.total_tax_liability)"
```

Then open the IRS table and do the arithmetic by hand:

```
https://www.irs.gov/pub/irs-drop/rp-24-40.pdf#page=6
Table 3, row "Over $48,475 but not over $103,350":
    5,578.50 + 0.22 × (taxable income - 48,475)
```

And check the deduction the software used against enacted law:

```
https://www.irs.gov/forms-pubs/how-to-update-withholding-to-account-for-tax-law-changes-for-2025
section "An increase in the standard deduction"
```

Exhibit source and captures: docs/tie-out/withholding-2026-09-06/. The case is synthetic; no client data appears in this document.